

Shama-46
403

Maria
Eco-46

Autocorrelation: what happens if the error terms are correlated.

The Nature of autocorrelation

The term autocorrelation may be defined as correlation between members of series of observations ordered in time or space.

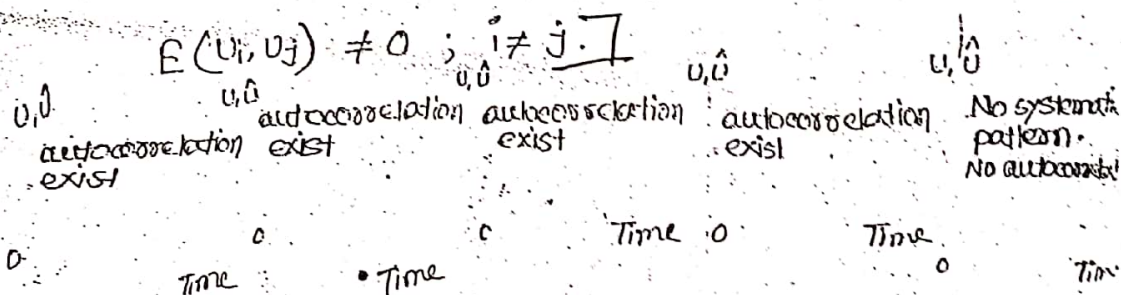
In the regression context, the classical linear regression model assumes that such autocorrelation does not exist in the disturbances u_i .
Symbolically,

$$E(u_i, u_j) = 0 ; i \neq j$$

which means the disturbance term relating to any observations is not influenced by the disturbance term relating to any other.

If autocorrelation exist then we can write,

$$E(u_i, u_j) \neq 0 ; i \neq j$$



Q Why autocorrelation considered as a problem ?

Under the autocorrelation the usual OLS estimators, although linear, unbiased and asymptotically normally distributed are no longer minimum variance among all linear unbiased estimators.

In short, they are not efficient relative to other linear and unbiased estimators.

put differently, they may not be BLUE. As a result the usual t , F and χ^2 test may not be valid.

Page / 43

G-1a part)

Autocorrelation Versus Serial Correlation:

Autocorrelation is related to lag correlation of a given series with itself, lagged by a number of time units.

Example:

$U_1, U_2, U_3, \dots, U_{n-1}, \dots, U_n$ (1)

$U_2, U_3, U_4, \dots, U_n$ (2)

Here series (1) is the series (2)'s lagged by one time period.

Serial correlation is related to lag correlation between two different series.

Example:

$U_1, U_2, U_3, \dots, U_n$

$V_1, V_2, V_3, \dots, V_n$

Here, U and V are two different time series.

Q Why does autocorrelation occur?

There are several reasons, some of which are as follows:

1) Inertia: (निष्क्रियता) 439 Page

A salient ^(main) feature of most economic time series is inertia or sluggishness.

For example, GNP, Price indexes, production, employment and unemployment cycles.

Because of inertia, starting at the bottom of the recession, when economic recovery starts, most of these series start moving upward and it continues until something happens to slow them down.

Thus in regressions involving time series data successive observations are likely to be interdependent.

2) Specification bias (Excluded variable case):

Sometimes there are some variables that were originally candidates but were not included in the model for variety of reasons should be included.

This is the case of excluded variable specification bias.

For example, we have the following demand model,

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + \beta_4 X_{4t} + U_t \quad \text{--- (1)}$$

Here, Y = quantity of beef demanded

X_2 = price of beef

X_3 = Consumer income

X_4 = price of pork

t = time.

For some reason if we run the following regression:

$$Y_t = \beta_1 + \beta_2 X_{2t} + \beta_3 X_{3t} + V_t \quad \text{--- (2)}$$

If equation ① is true model and to the extent the price of pork affects the consumption of beef, v will reflect a systematic pattern, thus creating autocorrelation.

Often the inclusion of such variables removes the correlation pattern.

3. Specification bias : Incorrect functional form

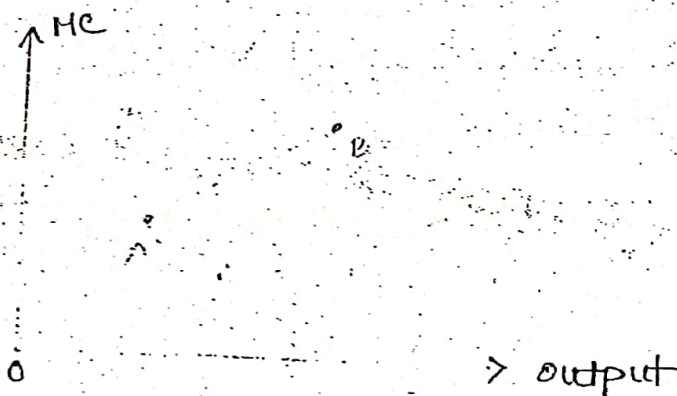
Suppose, we have the cost-output model,

$$\text{Marginal cost}_i = \beta_1 + \beta_2 \cdot \text{output}_i + \beta_3 \cdot \text{output}_i^2 + U_i \quad \text{--- ①}$$

But if we fit the following model -

$$\text{Marginal cost}_i = \alpha_1 + \alpha_2 \cdot \text{output} + V_i \quad \text{--- ②}$$

These two marginal cost curves are shown by the figure -



Between point A and B the linear marginal cost curve overestimate the true MC curve. otherwise vice-versa.

This is to be expected because there is an effect of the $(\text{output})^2$ term on marginal cost.

Thus, V_i will reflect autocorrelation because of using an incorrect functional form.

Q4. Cobweb phenomenon:

Let the supply function of the farmers in a given period is -

$$\text{Supply}_t = \beta_1 + \beta_2 P_{t-1} + U_t$$

Suppose at the end of period t , price P_t turns out to be lower than P_{t-1} .

Therefore in period $t+1$ farmers may very well decide to produce less than they did in period t .

The disturbances U_t are autocorrelated use if the farmers over produce in

in period t , they are likely to reduce their production in $t+1$ and so on, leading to a cobweb pattern.

5. Lags:

In time series regression, atleast one of the explanatory variables is the lagged value of the dependent variable. Such regression is known as autoregression.

For example, consumption expenditure depends not only on income but also on other variables. Now consider the following model.

$$\text{Consumption}_t = \beta_1 + \beta_2 \cdot \text{income} + \beta_3 \cdot \text{Consumption}_{t-1} + U_t$$

Now if we neglect the lagged term then U_t reflect autocorrelation.

6. Manipulation of Data:

In empirical analysis, the raw data are often manipulated. For example, in time series regressions involving quarterly data, such data are usually derived from

the monthly data by simply adding three monthly observations and dividing the sum by 3.

This averaging introduces smoothness into the data by dampening the fluctuations in the monthly data.

Therefore the graph plotting the quarterly data looks much smoother than the monthly data. and this smoothness may itself lend to a systematic pattern in the disturbances, thereby introducing autocorrelation.

7. Data Transformation:

Consider the following model:

$$Y_t = \beta_1 + \beta_2 X_t + u_t \quad \text{--- (1)}$$

where, Y = consumption expenditure

X = income.

Since equation (1) holds true at every time period, it holds true also in the previous time period, $(t-1)$. so we can write,

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + u_{t-1} \quad \text{--- (2)}$$

Now if we subtract equation (2) from (1) then,

$$\Delta Y_t = \beta_2 \Delta X_t + V_t \quad \text{--- (3)}$$

Where,

$$\Delta Y_t = Y_t - Y_{t-1}, \quad \Delta X_t = X_t - X_{t-1}, \quad V_t = U_t - U_{t-1}$$

Now if the error term in (1) satisfies the standard OLS assumptions, particularly the assumption of no autocorrelation, it can be shown that error term V_t is autocorrelated.

The point of the preceding example is that sometimes autocorrelation may be induced as a result of transforming the original model.

□ How the disturbance is generated?

Assume that,

$$U_t = \rho U_{t-1} + E_t \quad ; \quad -1 < \rho < 1 \quad \text{--- (1)}$$

where,

ρ = coefficient of autocovariance.

E_t is the stochastic disturbance term such that it satisfies the standard OLS assumptions, namely,

$$E(E_t) = 0$$

$$\text{Var}(E_t) = \sigma_e^2$$

$$\text{cov}(E_t, E_{t+s}) = 0, \quad s \neq 0$$

Equation (1) states that the value of the disturbance term in period t is equal to ρ times its value in the previous period plus a purely random error term.

The scheme in (1) is known as Markov 1st order autoregressive scheme which is known as AR(1).

The name autoregressive is appropriate because equation (1) can be interpreted as the regression of U_t on itself lagged one period.

It is first order because U_t and its immediate past value are involved, that is, the maximum lag is 1.

So the second order autoregressive scheme (AR_2) can be written as,

$$U_t = f_1 U_{t-1} + f_2 U_{t-2} + \epsilon_t \quad \text{--- (2)}$$

The coefficient of covariance in equation (1) can also be interpreted as the first-order coefficient of autocorrelation or more accurately, the coefficient of the autocorrelation at lag 1.

By definition, the coefficient of correlation between U_t and U_{t-1} is,

$$f = \frac{E\{[U_t - E(U_t)][U_{t-1} - E(U_{t-1})]\}}{\sqrt{\text{var}(U_t)} \sqrt{\text{var}(U_{t-1})}}$$

$$\rho = \frac{E(U_t U_{t-1})}{\text{Var}(U_{t-1})}$$

Since $E(U_t) = 0$ for each t and $\text{Var}(U_t) = \text{Var}(U_{t-1})$ because we are retaining the assumption of homoscedasticity.

✓ ☐ How is the variance under AR(1) scheme?

Assume that,

the two variable regression model is,

$$Y_t = \beta_1 + \beta_2 X_t + U_t$$

The OLS estimator of the slope coefficient is

$$\hat{\beta}_2 = \frac{\sum X_t Y_t}{\sum X_t^2}$$

$$\text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum X_t^2}$$

Under AR(1) scheme the variance of $\hat{\beta}_2$ is,

$$\sigma^2 (\hat{\beta}_2)_{AR(1)} = \frac{\sigma^2 \sum X_t^2}{\sum X_t^2} = \sigma^2$$

That is, $\text{var}(\hat{\beta}_2)_{AR1}$ depends on the $\text{var}(\hat{\beta}_2)$ and the sample autocorrelation between the values taken by the regressor X at various lags.

If $\rho = 0$ then,

$$\text{var}(\hat{\beta}_2) = \text{var}(\hat{\beta}_2)_{AR1}.$$

→ The BLUE estimator in the presence of autocorrelation:

By using generalized least square (GLS) method,
the two variable model under AR(1) scheme,
the BLUE estimator of β_2 is -

$$\hat{\beta}_{2GLS} = \frac{\sum_{t=2}^n (X_t - \rho X_{t-1})(Y_t - \rho Y_{t-1})}{\sum_{t=2}^n (X_t - \rho X_{t-1})^2} + C$$

Here, $C = \text{correlation factor}$.

And,

$$\text{var}(\hat{\beta}_2)_{GLS} = \frac{\rho^2}{\sum_{t=2}^n (X_t - \rho X_{t-1})^2} + D$$

Here, D is also correction factor.

The GLS estimator of β_2 incorporates the autocorrelation parameter ρ , whereas the OLS estimator of β_2 simply neglects it.

That is why GLS estimator is BLUE and not the OLS estimator.

In short, under autocorrelation, the GLS estimator is BLUE and the minimum variance is $\text{var}(\beta_2)_{\text{GLS}}$.

✓

✓ Consequences of using OLS in the presence of Autocorrelation:

As in the case of heteroscedasticity, in the presence of autocorrelation, the OLS estimators are still linear unbiased as well as consistent and asymptotically normally distributed, but they are not efficient because their variance is not minimum.

When we use the OLS method, there happens two cases in the hypothesis testing:

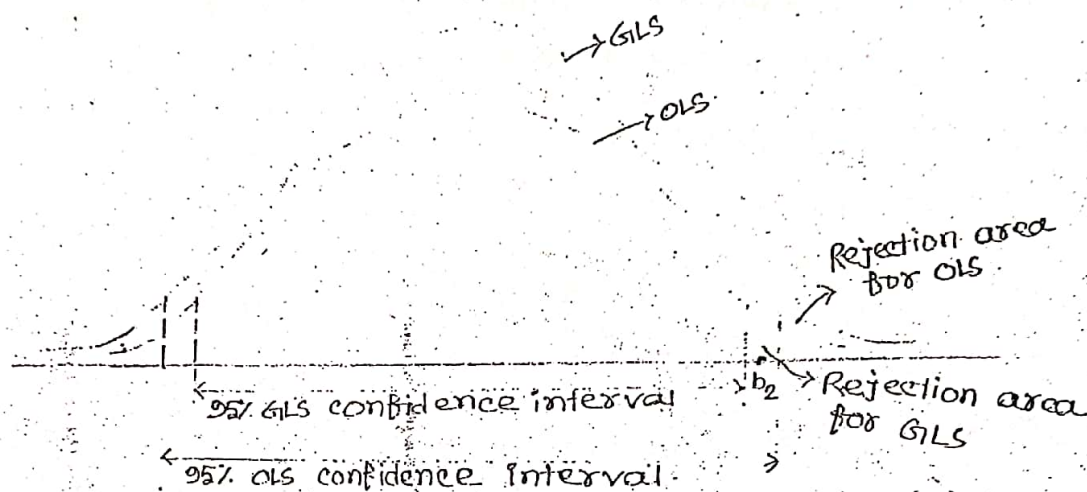
✓ Case-1: OLS estimation allowing for autocorrelation

In this case β_2 is not blue. It is not asymptotically efficient. The implication of this finding for hypothesis testing is clear:

The coefficient is statistically insignificant even though in fact it may be. If we use GLS procedure then it may be significant.

The difference can be seen from the

Following figure:



Here, we show the 95% OLS and GLS confidence interval.

The hypothesis is:

$$H_0: \text{True } \beta_2 = 0.$$

Consider, a particular estimate of $\beta_2 = b_2$.
If we use OLS confidence interval, we could accept the hypothesis that true $\beta_2 = 0$ with 95% confidence.

But if we were to use GLS confidence interval, we could reject the null hypothesis that true

$\beta_2 = 0$, for b_2 lies in the region of rejection.

The message is: To establish confidence intervals and to test hypothesis, one should use GLS and not OLS even though the estimators derived from OLS are unbiased and consistent.

Case-2: OLS estimation disregarding autocorrelation

We find serious problem if we mistakenly believe that the usual assumption of the classical model hold true, that is, we disregard the problem of autocorrelation.

The problem will arise for the following reasons:

- i) The residual variance, $\hat{\sigma}^2 = \frac{\sum \hat{u}_t^2}{n-2}$ is likely to underestimate the true σ^2 .
- ii) As a result we are likely to overestimate R^2 .
- iii) Even if σ^2 is not underestimated, $\text{var}(\hat{\beta}_2)$ may underestimate $\text{var}(\hat{\beta}_2)_{\text{AR}(1)}$, even though $\text{var}(\hat{\beta}_2)_{\text{AR}(1)}$ is inefficient compared to $\text{var}(\hat{\beta}_2)_{\text{OLS}}$.
- iv) Therefore, the usual t and F tests of significance are no longer valid and if applied, are likely to give seriously misleading conclusion about the

statistical significance of the estimated regression coefficients.

To establish some of these propositions, assume two variable model from which we can write,

$$\hat{\sigma}^2 = \frac{\sum \hat{u}_t^2}{n-2}$$

It provides an unbiased estimator of σ^2 that is $E(\hat{\sigma}^2) = \sigma^2$

But if there is autocorrelation, it can be shown that,

$$E(\hat{\sigma}^2) = \frac{\sigma^2 \left\{ n - \left(\frac{2}{1-\rho} \right) \right\} - 2\rho\sigma^2}{n-2} \quad \text{--- (1)}$$

where,

$$\rho = \frac{\sum_{t=1}^{n-1} X_t X_{t-1}}{\sum_{t=1}^n X_t^2}$$

If ρ and n are both positive it is apparent from equation (1) that $E(\hat{\sigma}^2) < \sigma^2$ that is the usual residual variance will underestimate the true σ^2 .

In other words, $\hat{q}^2$ will be biased downward. So this bias in $\hat{q}^2$ will be transmitted to $\text{var}(\hat{\beta}_2)$. Because,

$$\text{var}(\hat{\beta}_2) = \frac{\hat{q}^2}{\sum X_t^2}$$

But even if $\hat{q}^2$ is not underestimated, $\text{var}(\hat{\beta}_2)$ is a biased estimator of $\text{var}(\hat{\beta}_2)_{AR(1)}$.

If ρ is positive and the X 's are positively correlated, then it is clear that,

$$\text{var}(\hat{\beta}_2) < \text{var}(\hat{\beta}_2)_{AR(1)}$$

Therefore, if we use $\text{var}(\hat{\beta}_2)$, we shall underestimate the standard error of the estimator $\hat{\beta}_2$.

As a result the t ratio as, $t = \frac{\hat{\beta}_2}{\text{se}(\hat{\beta}_2)}$ is overestimated.

Hence we can not reach a valid result.

☐ Detecting auto correlation:

There are various ways of detecting autocorrelation:

1) Durbin Watson d test:

See previous chapter.

2) A General Test of Autocorrelation:

The Breusch-Godfrey (BG) Test

To avoid some of the pitfalls of Durbin-Watson d test of autocorrelation, Breusch and Godfrey have developed a test known as BG test. It is also known as LM test.

This test follows:

1) Nonstochastic regressors, such as the lagged values of the regressand.

2) It allows higher-order autoregressive scheme, such as $AR(1)$, $AR(2)$ etc.

- 3) It allows simple or higher order moving average of white noise error term such as ϵ_t .

Consider two variable model:

$$Y_t = \beta_1 + \beta_2 X_t + u_t \quad \text{--- (1)}$$

Say, u_t follows the p th order autoregressive AR(p) scheme as,

$$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \dots + \rho_p u_{t-p} + \epsilon_t \quad \text{--- (2)}$$

The null hypothesis is that,

$$H_0: \rho_1 = \rho_2 = \dots = \rho_p = 0 \quad \text{--- (3)}$$

That is, there is no serial correlation of any order.

The BG test involves the following steps:

- 1) Estimate equation (1) by OLS and obtain the residuals $\hat{u}_t$.
- 2) Regress $\hat{u}_t$ on the original X_t (if there is more than one X variable in the original model, include them also) and $u_{t-1}, u_{t-2}, \dots, u_{t-p}$,

where these are the lagged values of the estimated residuals, $\hat{U}_t$.

Now we can run the following regression:

$$\hat{U}_t = \alpha_2 + \alpha_2 X_t + \hat{f}_1 \hat{U}_{t-1} + \hat{f}_2 \hat{U}_{t-2} + \dots + \hat{f}_p \hat{U}_{t-p} + \epsilon_t$$

and obtain R^2 from this regression.

3) If the sample size is large ($n \rightarrow \infty$), BG test shows that,

$$(n-p)R^2 \approx \chi_p^2 ; df = p$$

If $(n-p)R^2$ exceeds the critical χ^2 at a level of significance, we reject the null hypothesis: at least one f is statistically different from zero. and this is the evidence of autocorrelation.

Ex Correcting for autocorrelation:

The method of Generalized least squares (GLS)

The remedy depends on the knowledge about the structure of autocorrelation.

Consider the two-variable regression model:

$$Y_t = \beta_1 + \beta_2 X_t + U_t \quad \text{--- (1)}$$

and the error term follows the AR(1) scheme.

$$U_t = \rho U_{t-1} + \varepsilon_t \quad \text{--- (2)} ; -1 < \rho < 1$$

We will now consider the following two cases:

1) When ρ is known:

As the coefficient of first order autocorrelation is known, the problem of autocorrelation can be easily solved.

As equation (1) holds true at time t , it also holds true at time $(t-1)$. Hence,

$$Y_{t-1} = \beta_1 + \beta_2 X_{t-1} + U_{t-1} \quad \text{--- (3)}$$

Multiplying (3) by ρ on both sides, we obtain,

$$\rho Y_{t-1} = \rho \beta_1 + \rho \beta_2 X_{t-1} + \rho U_{t-1} \quad \text{--- (4)}$$

Subtracting ④ from ① we get-

$$Y_t - \rho Y_{t-1} = \beta_1(1-\rho) + \beta_2(X_t - \rho X_{t-1}) + U_t - \rho U_{t-1} \quad \text{--- (5)}$$

$$\Rightarrow Y_t^* = \beta_1^* + \beta_2^* X_t^* + \varepsilon_t$$

Here,

$$Y_t^* = Y_t - \rho Y_{t-1}$$

$$\beta_1^* = \beta_1(1-\rho)$$

$$\beta_2^* = \beta_2$$

$$X_t^* = X_t - \rho X_{t-1}$$

$$\varepsilon_t = U_t - \rho U_{t-1}$$

This error term satisfies the usual OLS assumptions.

GLS is nothing but OLS applied to the transformed model that satisfied all the optimum properties, namely BLUE.

The equation ⑤ is known as generalized or quasi-difference equation.

It involves regressing y on x , not in the original form but in the difference form, which is obtained by subtracting a proportion $(=p)$ of the value of a variable in the previous time period from its value in the current time period.

In this differencing procedure we lose one observation because the first observation has no antecedent.

To avoid this problem, the first observation on y and x is transformed as,

$$Y_1 \sqrt{1-p^2}, \quad X_1 \sqrt{1-p^2}.$$

This transformation is known as the Prais-Winsten transformation.

2. When ρ is not known:

The first difference method

Since ρ lies between 0 and ± 1 , there are two extreme positions.

i) $\rho = 0$, that is, no serial correlation.

ii) $\rho = \pm 1$, that is, perfectly positive or negative autocorrelation.

Let assume,

$$\rho = +1$$

Equation (5) become,

$$Y_t - Y_{t-1} = \beta_2 (X_t - X_{t-1}) + U_t - U_{t-1}$$

$$\Rightarrow \Delta Y_t = \beta_2 \Delta X_t + \varepsilon_t \quad \text{--- (6)}$$

Here, Δ is the first difference operator.

The error term is free from serial correlation.

Now we can run regression on this first differences.

If the durbin-watson d is quite low then there exist autocorrelation.

An important feature of the first difference model, (6), is that there is no intercept in it.

But when time is important that is in time series there must include a time trend variable.

$$Y_t = \beta_1 + \beta_2 X_t + \beta_3 t + \epsilon_t$$

In this situation the first difference model

is,

$$\Delta Y_t = \beta_2 \Delta X_t + \beta_3 + \epsilon_t$$

Here exist an intercept, which is nothing but the coefficient of the trend variable.

Thus, when there present the intercept term in the first difference model then there must be a trend variable in the original model.

Again assume, $\rho = -1$

So,

$$\Delta Y_t = Y_t + Y_{t-1}$$

$$= 2\beta_1 + \beta_2(X_t + X_{t-1}) + U_t + U_{t-1}$$

$$\Rightarrow \frac{Y_t + Y_{t-1}}{2} = \beta_1 + \beta_2\left(\frac{X_t + X_{t-1}}{2}\right) + \frac{U_t + U_{t-1}}{2}$$

It is another type of transformation, we can say this model as moving average model.

The first difference transformation is valid only if $\rho = 1$.

There is a test, called the Breusch-Pagan-Ljung-Box test, to test the hypothesis that $\rho = 1$. It is called the g statistic.

$$g = \frac{\sum_{t=1}^n \hat{e}_t^2}{\sum_{t=1}^n U_t^2}$$

Here,

$\hat{e}_t$ = the OLS residuals from the first difference regression.

$\hat{u}_t$ = the OLS residuals from the original regression.

To test the significance of the ρ statistic assuming that the level form regression contains the intercept term, we can use the Durbin-Watson tables except that now the null hypothesis is that $\rho = 1$ rather than the durbin-watson hypothesis that $\rho = 0$.